Phase 2 Modeling Summary

Phase 2: Full-dataset modeling summary

Dataset: all 2,880 rows

Validation: 70/30 stratified holdout + repeated grouped CV on training split

Feature sets compared: embeddings only vs embeddings plus region and Dynamic World context

Current implementation differs from the original proposal: it uses the full dataset and a LightGBM ablation rather than the earlier 720-row subset plus ridge baseline plan.

Best held-out feature set by target:

- S1_VH: Embeddings + region + Dynamic World with R2=0.943, RMSE=1.175, MAE=0.858, r=0.971

- S1_VV: Embeddings + region + Dynamic World with R2=0.902, RMSE=1.260, MAE=0.824, r=0.951

- S1_VV_div_VH: Embeddings only with R2=0.707, RMSE=1.139, MAE=0.796, r=0.841

Interpretation:

- S1_VV is best with Embeddings + region + Dynamic World.

- S1_VH is best with Embeddings + region + Dynamic World.

- VV/VH ratio is best with Embeddings only.

- Mean grouped-CV R2 across the selected models ranges from 0.729 to 0.939.

- Important caveat: the outer holdout is stratified random within sampled regions, not spatially disjoint. Treat these as within-region generalization estimates rather than strict geographic transfer performance.

Direct vs structural VV-VH finding:

Phase 2 Modeling Summary (cont.)

- Using the saved LightGBM predictions, direct prediction of the stored dB-space target S1_VV_div_VH beats the structural baseline VV_hat - VH_hat for both feature sets.

- Embeddings only: direct R2=0.707, RMSE=1.139, MAE=0.796 vs structural R2=0.668, RMSE=1.212, MAE=0.863.

- Embeddings + region + Dynamic World: direct R2=0.699, RMSE=1.154, MAE=0.798 vs structural R2=0.677, RMSE=1.196, MAE=0.838.

Derived linear VV/VH ratio baselines:

- The stored S1_VV_div_VH target is a dB-space difference, so the new baseline reconstructs the physical VV/VH ratio from VV and VH before fitting/evaluation.

- Best overall linear-ratio baseline: LightGBM direct ratio (Embeddings only) with R2=0.584, RMSE=3.341, MAE=1.502, r=0.765.

- Ridge direct ratio best result: Ridge direct ratio (Embeddings + region + Dynamic World) (R2=0.534).

- Ridge log-ratio best result: Ridge log-ratio (Embeddings + region + Dynamic World) (R2=0.546).

- LightGBM direct ratio best result: LightGBM direct ratio (Embeddings only) (R2=0.584).

Held-out Performance by Feature Set

target	feature_set	cv_r2_mean	cv_rmse_mean	stability_r2_mean	stability_rmse_mean	r2	rmse	mae	pearson_r
S1_VH	Embeddings + region + Dynamic World	0.939	1.220	0.938	1.229	0.943	1.175	0.858	0.971
S1_VH	Embeddings only	0.940	1.211	0.939	1.223	0.941	1.194	0.861	0.970
S1_VV	Embeddings + region + Dynamic World	0.908	1.162	0.907	1.163	0.902	1.260	0.824	0.951
S1_VV	Embeddings only	0.910	1.143	0.910	1.145	0.900	1.272	0.834	0.949
S1_VV_div_VH	Embeddings only	0.731	1.057	0.729	1.060	0.707	1.139	0.796	0.841
S1_VV_div_VH	Embeddings + region + Dynamic World	0.732	1.056	0.730	1.059	0.699	1.154	0.798	0.836

The holdout test set is shared across the two feature-set ablations. It is a stratified random holdout within sampled regions rather than a spatially disjoint holdout, so these estimates may be optimistic relative to stricter geographic transfer tests.

Polarization Difference Experiments

Polarization Difference Experiments

This addendum compares direct and structural approaches for the stored polarization-difference target S1_VV_div_VH = S1_VV - S1_VH.

Best Ridge direct model: Ridge (VV-VH) [Embeddings + region + Dynamic World] with R2=0.708, RMSE=1.137, MAE=0.801.

Best LightGBM direct model: LightGBM (VV-VH) [Embeddings only] with R2=0.707, RMSE=1.139, MAE=0.796.

Best structural baseline: Structural baseline (VV_hat - VH_hat) [Embeddings + region + Dynamic World] with R2=0.708, RMSE=1.137, MAE=0.801.

Interpretation:

- Direct VV-VH prediction tests whether the embedding captures polarization contrast as its own target.

- The structural baseline tests whether separate VV and VH Ridge models preserve the difference after subtraction.

- A follow-up comparison using the saved LightGBM VV_hat and VH_hat predictions gives the same directional conclusion: direct VV-VH prediction remains better than reconstructing VV_hat - VH_hat.

- Embeddings only follow-up: direct R2=0.707, RMSE=1.139, MAE=0.796; structural R2=0.668, RMSE=1.212, MAE=0.863.

- Embeddings + region + Dynamic World follow-up: direct R2=0.699, RMSE=1.154, MAE=0.798; structural R2=0.677, RMSE=1.196, MAE=0.838.

Polarization Difference Comparison

feature_set	model_label	target	training_target	r2	rmse	mae	pearson_r
Embeddings only	Ridge (VH) [Embeddings only]	S1_VH	S1_VH	0.929	1.317	0.945	0.964
Embeddings only	Ridge (VV) [Embeddings only]	S1_VV	S1_VV	0.868	1.456	0.938	0.933
Embeddings only	Ridge (VV-VH) [Embeddings only]	S1_VV_div_VH	S1_VV_div_VH	0.708	1.137	0.801	0.842
Embeddings only	Structural baseline (VV_hat - VH_hat) [Embeddings only]	S1_VV_div_VH	S1_VV+S1_VH	0.708	1.137	0.801	0.842
Embeddings only	LightGBM (VV-VH) [Embeddings only]	S1_VV_div_VH	S1_VV_div_VH	0.707	1.139	0.796	0.841
Embeddings + region + Dynamic World	Ridge (VH) [Embeddings + region + Dynamic World]	S1_VH	S1_VH	0.929	1.315	0.944	0.964
Embeddings + region + Dynamic World	Ridge (VV) [Embeddings + region + Dynamic World]	S1_VV	S1_VV	0.870	1.445	0.934	0.934
Embeddings + region + Dynamic World	Ridge (VV-VH) [Embeddings + region + Dynamic World]	S1_VV_div_VH	S1_VV_div_VH	0.708	1.137	0.801	0.842
Embeddings + region + Dynamic World	Structural baseline (VV_hat - VH_hat) [Embeddings + region + Dynamic World]	S1_VV_div_VH	S1_VV+S1_VH	0.708	1.137	0.801	0.842
Embeddings + region + Dynamic World	LightGBM (VV-VH) [Embeddings + region + Dynamic World]	S1_VV_div_VH	S1_VV_div_VH	0.699	1.154	0.798	0.836

The structural baseline subtracts Ridge VV and Ridge VH predictions and evaluates the result against the stored dB-space polarization difference target.

Direct vs Structural Follow-up

feature_set	approach	r2	rmse	mae	prediction_file
Embeddings only	Direct LightGBM VV-VH	0.707	1.139	0.796	test_predictions_embedding_only_S1_VV_div_VH.csv
Embeddings only	Structural VV_hat - VH_hat	0.668	1.212	0.863	test_predictions_embedding_only_S1_VV_div_VH_structural_from_vv_minus_vh.csv
Embeddings + region + Dynamic World	Direct LightGBM VV-VH	0.699	1.154	0.798	test_predictions_embedding_plus_context_S1_VV_div_VH.csv
Embeddings + region + Dynamic World	Structural VV_hat - VH_hat	0.677	1.196	0.838	test_predictions_embedding_plus_context_S1_VV_div_VH_structural_from_vv_minus_vh.csv

These rows are computed from the saved LightGBM test predictions: direct uses the saved S1_VV_div_VH predictions, while structural reconstructs VV_hat - VH_hat from the saved VV and VH prediction files.

VV/VH Ratio Baselines

Phase 2 ratio-baseline addendum

These rows compare direct-ratio, log-ratio, and structural baselines on the derived linear VV/VH target.

The target is computed from the existing VV and VH columns and evaluated on the linear ratio scale.

Best overall baseline: LightGBM direct ratio (Embeddings only) with R2=0.584, RMSE=3.341, MAE=1.502, r=0.765.

Model families in this addendum:

- Ridge direct ratio: fit Ridge directly on VV/VH.

- Ridge log-ratio: fit Ridge on log(VV/VH), then exponentiate predictions.

- LightGBM direct ratio: fit LightGBM directly on VV/VH.

- Ridge structural ratio: predict VV and VH separately with Ridge, then reconstruct VV/VH.

Derived Linear VV/VH Baselines

feature_set	model_label	training_target	r2	rmse	mae	pearson_r
Embeddings only	LightGBM direct ratio (Embeddings only)	S1_VV_div_VH_linear	0.584	3.341	1.502	0.765
Embeddings only	Ridge log-ratio (Embeddings only)	S1_log_VV_div_VH	0.539	3.516	1.454	0.755
Embeddings only	Ridge structural ratio (Embeddings only)	S1_VV+S1_VH	0.539	3.516	1.454	0.755
Embeddings only	Ridge direct ratio (Embeddings only)	S1_VV_div_VH_linear	0.527	3.561	1.784	0.726
Embeddings + region + Dynamic World	LightGBM direct ratio (Embeddings + region + Dynamic World)	S1_VV_div_VH_linear	0.569	3.401	1.492	0.755
Embeddings + region + Dynamic World	Ridge log-ratio (Embeddings + region + Dynamic World)	S1_log_VV_div_VH	0.546	3.491	1.453	0.756
Embeddings + region + Dynamic World	Ridge structural ratio (Embeddings + region + Dynamic World)	S1_VV+S1_VH	0.546	3.491	1.453	0.756
Embeddings + region + Dynamic World	Ridge direct ratio (Embeddings + region + Dynamic World)	S1_VV_div_VH_linear	0.534	3.534	1.779	0.731

These baselines evaluate the physical linear VV/VH ratio reconstructed from the stored VV and VH columns. The dataset's original S1_VV_div_VH target remains the dB-space difference used in the main LightGBM tables above.

Repeated Grouped-CV Stability

target	feature_set	r2_mean	r2_std	rmse_mean	rmse_std	mae_mean	mae_std
S1_VH	Embeddings only	0.939	0.008	1.223	0.089	0.921	0.062
S1_VH	Embeddings + region + Dynamic World	0.938	0.006	1.229	0.069	0.925	0.047
S1_VV	Embeddings only	0.910	0.008	1.145	0.042	0.832	0.043
S1_VV	Embeddings + region + Dynamic World	0.907	0.006	1.163	0.010	0.843	0.030
S1_VV_div_VH	Embeddings only	0.729	0.029	1.060	0.021	0.787	0.020
S1_VV_div_VH	Embeddings + region + Dynamic World	0.730	0.027	1.059	0.028	0.785	0.021

Each row summarizes 12 grouped validation folds: 4 folds across 3 random seeds.

Regional S1_VV Diagnostics

region	feature_set	r2	rmse	mae	pearson_r	n_test
amazon_forest	Embeddings only	0.933	1.152	0.785	0.966	216
california_coast	Embeddings only	0.920	1.041	0.756	0.960	216
iowa_ag	Embeddings only	0.850	1.308	0.758	0.929	216
sf_bay_urban	Embeddings only	0.868	1.533	1.037	0.934	216
amazon_forest	Embeddings + region + Dynamic World	0.935	1.129	0.778	0.967	216
california_coast	Embeddings + region + Dynamic World	0.922	1.029	0.750	0.961	216
iowa_ag	Embeddings + region + Dynamic World	0.839	1.357	0.789	0.923	216
sf_bay_urban	Embeddings + region + Dynamic World	0.878	1.474	0.981	0.940	216

Region-level diagnostics on the held-out test set.

Land-use S1_VV Diagnostics

dw_label_name	feature_set	r2	rmse	mae	pearson_r	n_test
bare	Embeddings only	0.665	1.254	0.863	0.831	96
built	Embeddings only	0.734	2.098	1.226	0.871	96
crops	Embeddings only	0.770	1.005	0.777	0.878	96
flooded_vegetation	Embeddings only	0.924	0.777	0.545	0.962	96
grass	Embeddings only	0.857	0.980	0.716	0.928	96
shrub_and_scrub	Embeddings only	0.784	1.357	0.888	0.891	96
snow_and_ice	Embeddings only	0.891	1.047	0.767	0.945	96
trees	Embeddings only	0.859	0.901	0.724	0.928	96
water	Embeddings only	0.841	1.504	1.002	0.923	96
bare	Embeddings + region + Dynamic World	0.699	1.189	0.826	0.849	96
built	Embeddings + region + Dynamic World	0.732	2.107	1.233	0.872	96
crops	Embeddings + region + Dynamic World	0.782	0.978	0.724	0.885	96
flooded_vegetation	Embeddings + region + Dynamic World	0.917	0.810	0.559	0.958	96
grass	Embeddings + region + Dynamic World	0.858	0.978	0.691	0.929	96
shrub_and_scrub	Embeddings + region + Dynamic World	0.788	1.345	0.894	0.893	96
snow_and_ice	Embeddings + region + Dynamic World	0.898	1.016	0.761	0.948	96
trees	Embeddings + region + Dynamic World	0.847	0.938	0.744	0.923	96
water	Embeddings + region + Dynamic World	0.850	1.462	0.988	0.931	96

Land-use diagnostics on the held-out test set.

Feature Importance Summary

Top 5 features by permutation importance

Embeddings only

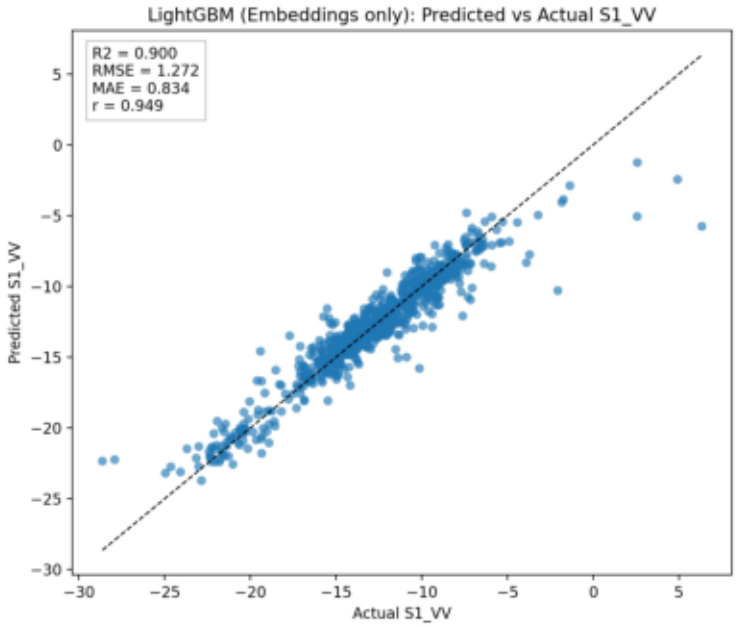
```
S1_VV: A25 (0.126), A27 (0.109), A49 (0.066), A17 (0.039), A19 (0.037)
S1_VH: A25 (0.268), A27 (0.066), A49 (0.030), A19 (0.028), A34 (0.023)
S1_VV_div_VH: A25 (0.126), A24 (0.047), A63 (0.042), A39 (0.035), A07 (0.033)
```

Embeddings + region + Dynamic World

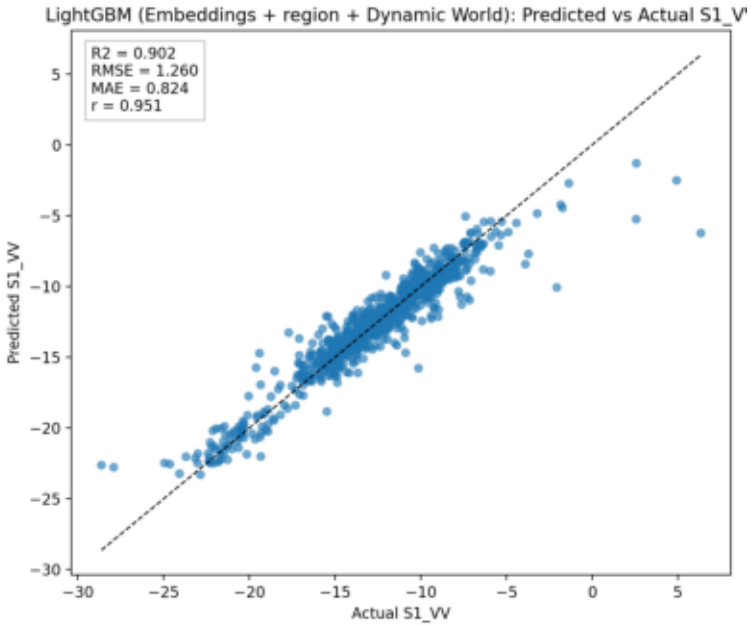
```
S1_VV: water (0.113), A27 (0.084), A49 (0.055), A25 (0.041), A17 (0.037)
S1_VH: water (0.117), A25 (0.105), A27 (0.045), bare (0.030), A49 (0.026)
S1_VV_div_VH: A25 (0.077), A31 (0.060), A39 (0.048), A63 (0.035), trees (0.028)
```

Predicted vs Actual

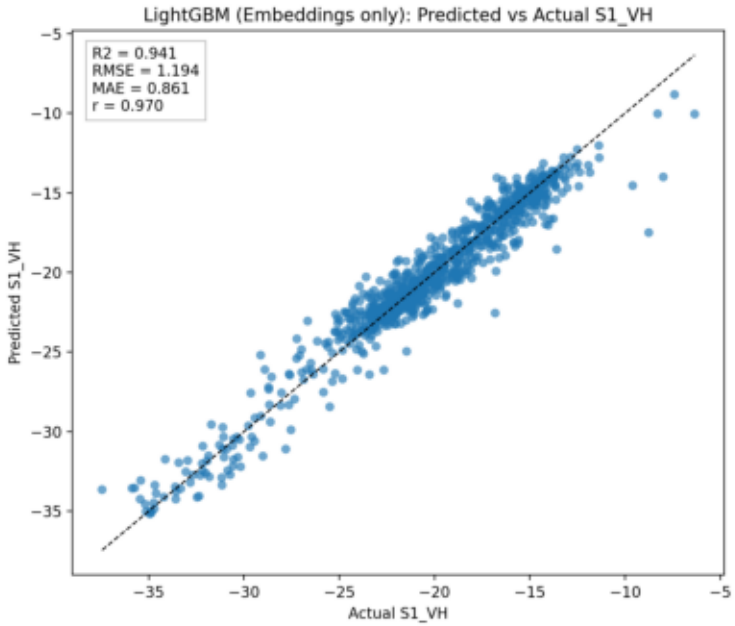
predicted vs actual embedding only S1 VV



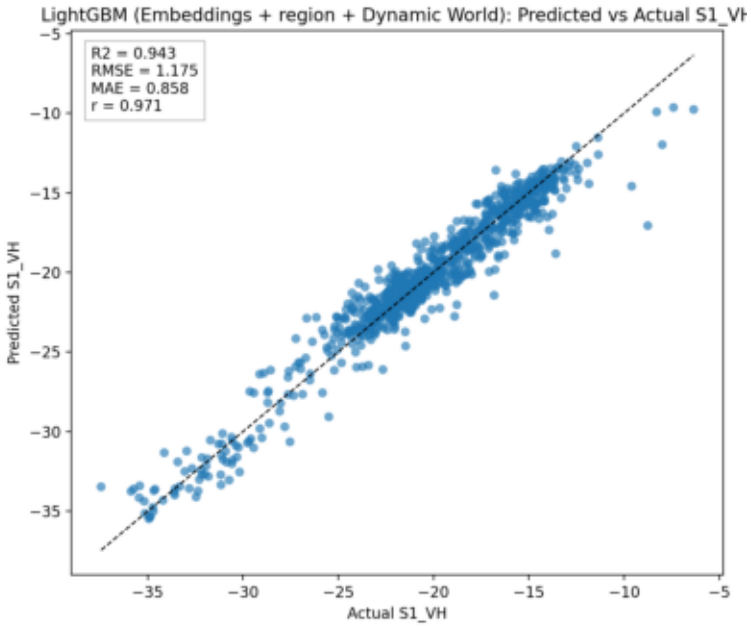
predicted vs actual embedding plus context S1 VV



predicted vs actual embedding only S1 VH

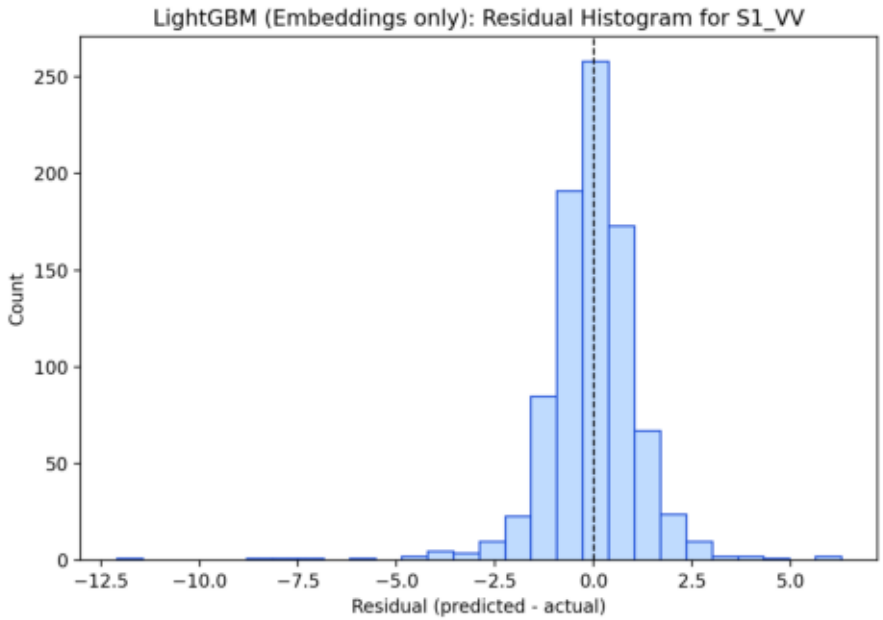


predicted vs actual embedding plus context S1 VH

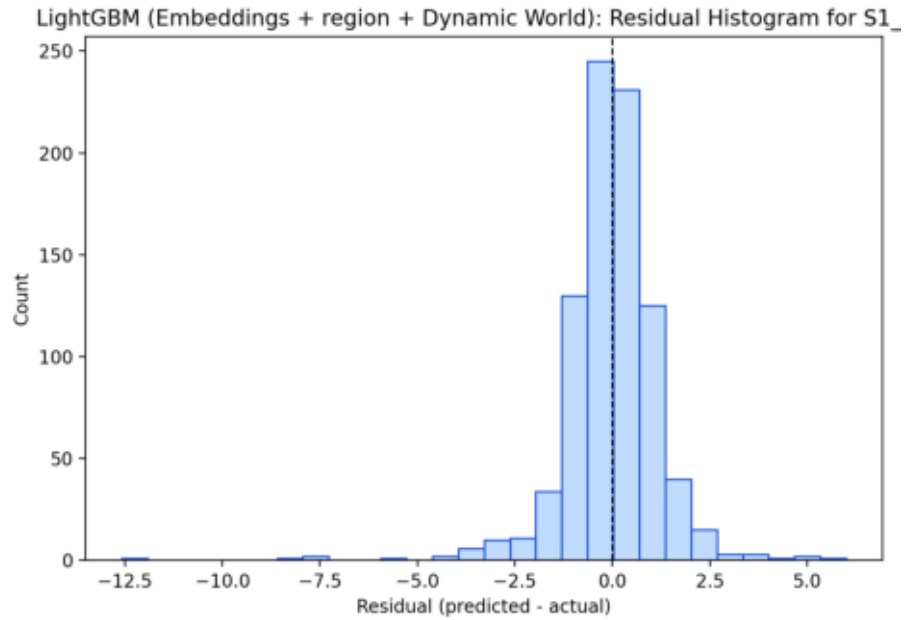


Residual Histograms

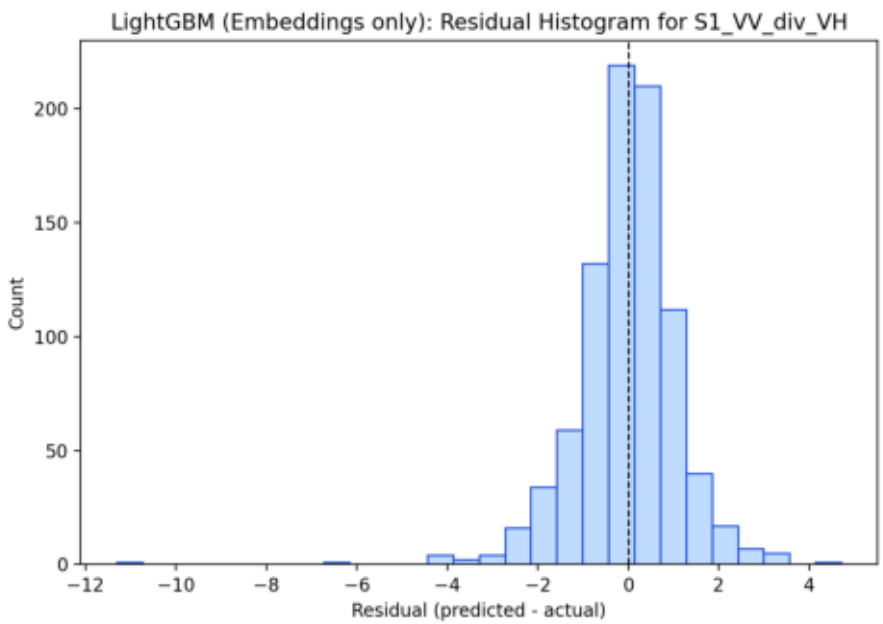
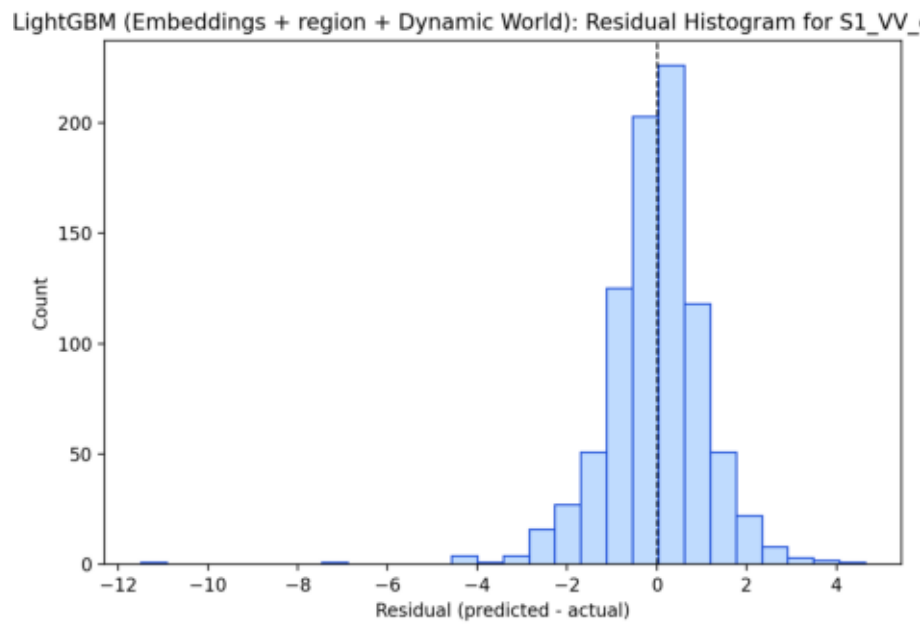
residual histogram embedding only S1 VV



residual histogram embedding plus context S1 VV



residual histogram embedding only S1 VV div VH



residual histogram embedding plus context S1 VV div VH